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The Effect of Retrieval Practice in Primary School Vocabulary Learning

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Summary: The testing effect refers to the finding that retrieval practice leads to better long-term retention than additional study of course material. In the present study, we examined whether this finding generalizes to primary school vocabulary learning. We also manipulated the word learning context. Children were introduced to 20 words by listening to a story in which novel words were embedded (story condition) or by listening to isolated words (word pairs condition). The children practised the meaning of 10 words by retrieval practice and 10 words by restudy. After 1 week, they completed a cued recall test and a multiple choice test. Words learned by retrieval practice were recalled better than words learned by additional study, but there was no difference in recognition. Furthermore, the children in the word pairs condition outperformed the children in the story condition. These results show that retrieval practice may improve vocabulary learning in children. Copyright © 2013 John Wiley & Sons, Ltd.

INTRODUCTION

Vocabulary learning has become one of the core components of language learning in the last 25 years (e.g. Vermeer, 2001). Already by the end of grade 2, large differences exist in children's vocabulary size. On average, children will then have acquired around 6000 word meanings, whereas children with a greater vocabulary knowledge have acquired around 8000 word meanings and children with a lower vocabulary knowledge just 4000 (Biemiller, 2005). Many studies have shown that vocabulary knowledge is a significant predictor of reading comprehension (e.g. Anderson & Freebody, 1981; Biemiller & Boote, 2006). Without sufficient knowledge of words, it is difficult or even impossible to understand a text. Furthermore, a smaller vocabulary size of children has been found to be persistent throughout the school years and can harm later school success (e.g. Baker, Simmons, & Kame'enui, 1998; Hart & Risley, 1995; Nation, 2001).

The differences in vocabulary knowledge and the problems that arise from insufficient vocabulary knowledge indicate that there is a need for vocabulary instruction that is optimal for learning new words and their meaning. Some of the characteristics of good vocabulary instruction have been described by Blachowicz and colleagues (Blachowicz, Fisher, Ogle, & Watts-Taffe, 2006). A central characteristic is repetition: words have to be practised during multiple exposures. The central question in the present study is *how* the words have to be practised during repeated study. In the literature, two kinds of repetition have frequently been investigated, namely *restudy* and *retrieval practice* (for a review, see Delaney, Verkoeijen, & Spirgel, 2010). More specifically, we wanted to investigate whether retrieval practice after initial learning can help children to learn new words more effectively than restudying. The instructional guideline to practise with retrieval is based on the so-called testing effect. The testing effect refers to the finding that retrieval practice enhances long-term retention more than additional study (for a review, see Roediger & Karpicke, 2006).

Most experiments on retrieval practice have been conducted using word lists or word pairs as study material (e.g. Carpenter, Pashler, & Vul, 2006; Carpenter, Pashler, Wixted, & Vul, 2008; Toppino & Cohen, 2009; Tulving, 1967; Wheeler, Ewers, & Buonanno, 2003). A classic example is an experiment by Webb (1921). In this study, participants studied 15 Hebrew words and their English equivalents for 5 min. For the next 3 min, half of the participants restudied the 15 word pairs, whereas the other participants received only the Hebrew words and had to retrieve their English equivalents (without getting feedback on their answers). In a similar cued recall test given 1 week later, participants in the retrieval practice condition recalled twice as many English equivalents as participants in the restudy condition, hence clearly demonstrating the benefits of retrieval practice for long-term recall performance.

Many studies have replicated these results in learning foreign vocabulary pairs (e.g. Carpenter et al., 2008; Carrier & Pashler, 1992; Karpicke, 2009; Karpicke & Roediger, 2008; Pashler, Cepeda, Wixted, & Rohrer, 2005; Pyc & Rawson, 2007, 2009, 2011; Toppino & Cohen, 2009) and in studies where people had to learn uncommon or infrequent words from their own language (e.g. Cull, 2000; Karpicke & Smith, 2012). For example, in the study of Karpicke and Smith (2012), participants studied 30 word pairs, each pair consisting of an uncommon English word and a one-word synonym. In the first part of the learning phase, all participants had to cycle through six study/recall periods in which they had to study and recall the word pairs during 7 s. When a participant correctly recalled the synonym of a word pair for the first time, the actual manipulation started. In the *drop* condition, the word pair was removed from the series of further study and recall trials. In the *repeated study* condition, the word pair was presented in two subsequent study trials. In the *repeated retrieval* condition, the word pair was presented in two subsequent retrieval trials. After 1 week, participants took a final cued recall test in which they had to give the correct synonym in response to a provided English cue word. The results of this test showed that the participants in the repeated retrieval condition outperformed the participants in the two other conditions.

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In sum, positive effects of repeated retrieval practice on long-term retention were found in both foreign vocabulary learning and first language learning. However, the described studies were conducted using only adult participants, and although the words were infrequent and uncommon, it was not clear whether participants had any prior knowledge about these words. Also, in most of these studies, words were studied without any context, whereas in primary school, new vocabulary words are always introduced within a meaningful context (e.g. Blachowicz et al., 2006). Therefore, the question remains whether the results from previous studies can be generalized to vocabulary learning in a primary school setting with young children.

A number of studies have reported positive effects of retrieval practice with primary school children. For example, Fritz, Morris, Nolan, and Singleton (2007) showed that preschoolers who learned names of toys recalled the names better after expanding retrieval practice than after expanding re-presentation or massed elaboration both after 1 min and after 1 day. Further, in a study by Rohrer, Taylor, and Sholar (2010), fourth and fifth graders had to learn regions or cities on map locations by retrieval practice and by restudying. On the final test after 1 day, they received both an identical task and a transfer task on the learned material. On both tasks, the children performed better after retrieval practice than after restudying. Marsh, Fazio, and Goswick (2012) showed that retrieval practice can benefit learning even for children of grade 2, but only when they received feedback. Finally, Bouwmeester and Verkoeijen (2011) compared restudying with taking an intervening free recall test with second to sixth graders learning Deese–Roediger–McDermott lists (Deese, 1959; Roediger & McDermott, 1995). One week later, the children were tested using a recognition test in which the children had to decide whether they had seen the word in the learning session 1 week before. In this recognition test, critical lures of the Deese–Roediger–McDermott lists were also included. The results showed an overall benefit of retrieval practice over restudying. Bouwmeester and Verkoeijen also found differences between children in how much they benefited from retrieval practice and that these differences were dependent on the amount of gist processing. Furthermore, the results showed no developmental trends in cognition; thus, the benefit of retrieval practice did not depend on the age of the children. In sum, these four studies showed benefits of retrieval practice with children, but these studies did not investigate primary school vocabulary learning in which new words have to be learned.

The central aim of the current study was to investigate whether retrieval practice is an effective strategy for improving vocabulary learning in primary school children. The first question was whether retrieval practice would indeed benefit long-term retention of new vocabulary words in primary school children. On the basis of the results from previous studies, our hypothesis was that retrieval practice after initial study would lead to better long-term retention of new vocabulary than additional study. A second question was whether presenting the words in a meaningful context would affect the hypothesized benefit of retrieval practice. In primary school vocabulary learning methods (e.g. Janssen & Van Ooijen, 2012; Van de Gein, Van de Guchte, & Kouwenberg, 2008), it is standard practice to present the words in a meaningful context. Therefore, we investigated whether retrieval practice is still effective in vocabulary learning in which a meaningful context is included. Earlier studies demonstrated that adding a (rich) context in itself can benefit vocabulary learning (for a review, see Stahl & Fairbanks, 1986). For example, in the study of Gipe (1979), four vocabulary learning methods were compared in third and fifth graders: an association method, a category method, a dictionary method and a context method. On the fill-in-the-blank test at the end of the week, the children in the context condition performed better than the children in the other conditions. These results indicate that the use of context is helpful in vocabulary learning. The present study evaluated whether the retrieval practice effect has practical value in the classroom, where providing a context in vocabulary learning is standard practice.

Context was manipulated in the current study in the following way. In the *story* condition, children were introduced to the words by reading a story that included the target words. This method is similar to the way in which vocabulary is typically introduced in the vocabulary learning method we used. In the *word pairs* condition, children were introduced to the words by reading a list of the target words without any context. Subsequently, in both conditions, half of the words was repeatedly restudied, whereas the other half of the words was repeatedly studied by retrieval practice. As dependent variables, children's long-term retention and recognition of the target words were measured after 1 week.

METHOD

Participants

Originally, 62 third graders of three Dutch primary schools participated. As one of the children missed the session in which the vocabulary size of the children was tested and one of the children missed the second session, the data from 60 children (28 boys, 32 girls) were used. The children were aged 8 to 11 years ($M = 9.24$ years, $SD = 0.49$). The primary schools were situated in an urban environment in Rotterdam, and the children had diverse ethnic backgrounds. The children knew they participated in an experiment, and their parents had given informed consent for participation.

The vocabulary size of the children was tested with a normed test for Dutch primary school children in grade 3 until grade 6 (Leeswoordenschattaak, Taaltoets Allochtone Kinderen Bovenbouw, Verhoeven & Vermeer, 1993). The test consisted of 50 sentences containing an underlined word. For each sentence, children had to select the best description of the underlined word from four options. The Cronbach's alpha of this test for third to sixth graders is sufficient to good for group-wise comparisons of mean performance ($\alpha = .81$, .83, .79 and .83, respectively). The participants in the two context conditions (story or word pairs) were matched on vocabulary size, resulting in 30 participants in the word pairs condition and 30 participants in the story condition. This procedure ensured that any differences between the two groups on the dependent variable were not caused by a priori differences in vocabulary size.

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Materials and design

A prose passage introducing 10 new words and a word list containing the synonyms of these 10 difficult words were selected from a grammar book typically used in grade 5. For the purpose of the experiment, the prose passage was adapted by deleting some irrelevant sentences and adding sentences with 10 other difficult words. In this way, the final prose passage and the final word list consisted of 20 difficult words. The synonyms of the words were not included in the prose passage. The difficulty level of the words was measured using the measure of lexical richness by computing the median word frequency (Schrooten & Vermeer, 1994; Vermeer, 2000). This median word frequency is a measure of the frequency of the word in several Dutch text books for primary school children (for a description, see Schrooten & Vermeer, 1994). Nineteen of the 20 words were in the list of Schrooten and Vermeer (1994). The median word frequency based on these 19 words was 3 (range 1 to 17), which is relatively low. The words were also pretested before the children started studying the words. Four children (6.7%) knew the synonyms of two words, and 10 children (16.7%) knew the synonym of one word. The other 46 children (76.7%) did not know any of the words. The Dutch words and synonyms and the English translations of the words and their synonyms are presented in Table 1.

In this experiment, we used a 2 (context) × 2 (learning condition) mixed design. The context (story or word pairs) was manipulated between subjects. Thirty of the children first listened to a story in which the words were introduced (story condition), whereas 30 of the children were introduced to the words by just listening to the words and their synonyms without any context (word pairs condition). In both conditions, the words were presented twice. The story is included in Dutch and in English in Appendices A and B, respectively. The learning condition was manipulated within subjects. After the introduction phase, half of the words was assigned to the restudy condition, in which the word pairs were studied seven times (SSSSSSS). The other half of the words was assigned to the retrieval practice condition, in which the word pairs were studied four times (e.g. *to weep – to cry*), retrieved once through cued recall (e.g. *to weep – …*), studied once more and then again retrieved (SSSSTST). In earlier studies, it was found that providing feedback after retrieval practice further strengthens the benefits of retrieval practice (e.g. McDaniel, Howard, & Einstein, 2009). Therefore, we also included a restudy phase in the retrieval practice condition, namely the final 'S' in the SSSTST sequence, as a way of providing indirect feedback to the children.

The assignment of the two lists of words to the learning conditions was counterbalanced. Also, the order in which the restudy and retrieval practice lists were alternated was counterbalanced. The order in which items were presented in the different learning phases was randomized anew for each phase. The dependent variables were long-term retention and long-term recognition, as measured by the scores on a cued recall test and the scores on a multiple choice test. The cued recall test was comparable with tests that are generally used in retrieval practice effect experiments on word-pair learning (e.g. Karpicke & Smith, 2012). In this test, the children had to orally retrieve the synonym of a given word (e.g. *to weep – …*). We scored the results on the final cued recall test by only awarding points to synonyms that were phonetically identical to the synonyms that the children had learned during the learning phases. For each answer, the children received either one point or no point. The multiple choice test consisted of a sentence in which one of the previously learned words was presented in a bold font along with four possible synonyms of the word (e.g. *Don't* ***weep*** *so much. – A. to pose; B. to cry; C. to talk; D. to throw*). On each test trial, one of the distractors was the synonym of another word that the children learned. For each test trail, children indicated which of the presented synonyms matched the bold word best. Both tests were administered after

Table 1. Dutch target words and synonyms and their English translations

Dutch target word	Dutch synonym	English target word	English synonym
apart	bijzonder	unique	special
baret	muts	a beret	a hat
beduusd	verrast	perplexed	confused
chaos	rommel	chaos	a mess
deponeren	gooien	to dispose	to throw
gift	cadeau	an offering	a present
heengaan	doodgaan	to pass away	to die
heimelijk	stiekem	secretly	sneaky
kris	mes	dagger	knife
kwiek	blij	briskly	happily
meedelen	vertellen	to describe	to tell
perplex	verbaasd	speechless	amazed
pronkstuk	mooi	showpiece	something beautiful
signaal	teken	a sign	a symbol
stug	stijf	rigid	firm
vaal	grauw	colourless	pale
vermoeden	idee	assumption	idea
verprutsen	verknoeien	to make a mess of	to fail
wenen	huilen	to weep	to cry
weerzinwekkend	lelijk	repulsive	ugly

Note. The English translations can deviate from the original Dutch meaning, making the synonyms in English seem to fit less well with the targets.

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1 week, which is a common delay at which long-term retention is measured in testing effect research (e.g. Roediger & Karpicke, 2006).

Procedure

The children were tested individually in a quiet room outside the classroom. The experiment consisted of two sessions. The first session was a learning session, followed by a test session 1 week later. The experiment was developed using E-Prime 2.0 and was presented on a laptop computer. The learning material was presented to the children on the screen, and the experimenter typed in the answers given by the children.

At the beginning of the learning session, the experimenter asked the children to give the meaning of each of the 20 new words and wrote down the answers of the children. After this pretest, the learning session consisted of seven phases. The first and second learning phases, in which the context was manipulated, took about 20 min altogether. The third to the seventh learning phase, in which the word pairs were practised, took about 15 min altogether.

In the first learning phase, the experimenter told the children that they would first listen to a story or to a word list depending on the context condition (story or word pairs) and that after this, they would be told the meaning of the words. Thus, at first, the experimenter read aloud the story or the word list depending on the context condition (story or word pairs) without any explanation of the words and without giving the synonyms of the words.

In the second learning phase, the experimenter read the story or the words to the children again, while the story (story condition) or the word list (word pairs condition) was presented on the screen. Thus, the children could read the story or the words while the experimenter was explaining the words. The experimenter gave the synonym of each word and explained its meaning by giving the synonym and a description of the word consisting of one sentence. Thus, in the story condition, after each sentence that contained a new word, the reading of the story was interrupted by the explanation of the new word.

In the third learning phase, the children in both conditions studied all 20 words. Every word and its synonym were presented for 8 s on the laptop (e.g. *to weep – to cry*), in random order. The children had to read the words and their synonyms aloud.

In the fourth learning phase, the children studied all 20 words again to practise the words sufficiently before the manipulation of learning condition (restudy or retrieval) would start. After this learning phase, there was a short break of 2 min in which the children had to work in a puzzle book. The puzzle book included, for example, math exercises, to divert the children from the language task.

In the fifth learning phase, the children first restudied half of the words, and then, they were tested on the other half, or the other way around, depending on the counterbalancing condition. The words in the restudy condition were studied together with their synonyms, as in the previous learning phase. The words in the retrieval practice condition were shown for 8 s, and the children were asked to read it aloud and then try to retrieve its synonym (e.g. *to weep – …*). After this phase, the children again had a short break of 2 min in which they continued with their puzzle book.

In the sixth learning phase, the children again restudied all 20 words once, with the same procedure as in the second learning phase. After this phase, the children again had a short break of 2 min in which they continued with their puzzle book.

In the seventh learning phase, the procedure was identical to the fifth learning phase. The children restudied the same 10 words and were tested on the same 10 words. After the children had finished this learning phase, they were asked not to talk about the words with their classmates, and they returned to their classroom.

In the test session, 1 week after the learning session, the children were given a cued recall test in which they had to orally retrieve the synonym of a given word. After that, the children completed a multiple choice test consisting of a list of written sentences, with each sentence containing (presented in bold font) a word they previously learned. The children were instructed to read aloud each sentence and to select out of four options the synonym that matched best with the bold word. If the children did not know the answer on this test, they had to guess which answer they considered best. These tests were administered individually in a quiet room outside the classroom. There were no time constraints. After the children had finished the tests, they were asked not to talk about the final test with their classmates, and they returned to their classroom.

RESULTS

First of all, we checked if our experimental groups were comparable with regard to vocabulary size by an independent sample *t*-test with the independent variable of learning condition (story or word pairs) and vocabulary size as a dependent variable for all 60 children. This analysis showed that there was no significant difference between the two context conditions on the vocabulary size measure, namely $t(58) = -1.12$, $p = .266$, $d = 0.29$. This means that our matching procedure, on the basis of the vocabulary size of the participants, was appropriate. Further analyses showed that vocabulary size correlated positively significant with the score on the restudied items (0.60) and on the retrieved items (0.56) of the cued recall test and also with the score on the restudied items (0.42) and on the retrieved items (0.47) of the multiple choice test. Thus, the higher the score on the vocabulary size test, the higher the score on the cued recall test and multiple choice test. Therefore, we used the *z*-score of this vocabulary size measure as a covariate in our analyses.

Learning session data

During the learning session, there were two phases in which the children received a practice test on half of the words. Table 2 shows the mean scores on these tests for each context group. We analysed these scores using a 2×2 mixed analysis of covariance (ANCOVA) with context as a between-subjects factor, retrieval practice phase as a within-subjects factor and the *z*-score of vocabulary size as a covariate.

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Table 2. Proportion correct on the initial tests in the first and second retrieval practice phase (*SD* in parentheses)

	Condition	
Retrieval practice phase	Word pairs ($n = 30$)	Story ($n = 30$)
First phase	0.52 (0.22)	0.41 (0.22)
Second phase	0.65 (0.22)	0.52 (0.22)

The mixed design ANCOVA on the practice tests during the learning sessions showed that the covariate, vocabulary size, was significantly related to the score on the two practice tests, $F(1, 57) = 55.19$, $p < .001$, $\eta^2 = 0.43$. With the covariate included in the model, there was a significant effect of retrieval practice phase, $F(1, 57) = 53.02$, $p < .001$, $\eta^2 = 0.48$. Children recalled more synonyms during the second retrieval practice phase than during the first retrieval practice phase (first: $M = 0.46$, $SD = 0.23$; versus second: $M = 0.58$, $SD = 0.23$). There was no significant interaction between retrieval practice phase and vocabulary size, $F < 1$. Also, there was a significant main effect of context, $F(1, 57) = 16.20$, $p < .001$, $\eta^2 = 0.13$, indicating that children in the word pairs condition recalled more synonyms in the retrieval practice phases than children in the story condition (word pairs: $M = 0.58$, $SD = 0.21$; versus story: $M = 0.47$, $SD = 0.21$). There was no significant interaction between retrieval practice phase and context, $F < 1$.[1]

Test session data

Table 3 shows the results on the final cued recall test, and Table 4 shows the results on the final multiple choice test. We analysed the results on both tests using a 2×2 mixed analysis of covariance (ANCOVA) with context as a between-subjects factor, learning condition as a within-subjects factor and *z*-score of vocabulary size as a covariate.

The mixed design ANCOVA on the cued recall test showed that the covariate, the vocabulary test, was significantly related to the score on the cued recall test, $F(1, 57) = 48.36$, $p < .001$, $\eta^2 = 0.42$. With the covariate included in the model, there was a significant effect of learning condition, $F(1, 57) = 15.03$, $p < .001$, $\eta^2 = 0.20$. There was no significant interaction between learning condition and vocabulary size, $F < 1$. There was a significant main effect of context, $F(1, 57) = 10.25$, $p = .002$, $\eta^2 = 0.09$, indicating that children in the word pairs condition recalled more synonyms than children in the story condition (word pairs: $M = 0.47$, $SD = 0.21$; versus story: $M = 0.39$, $SD = 0.19$). There was a marginally significant interaction between learning condition and context after controlling for the effect of the vocabulary test, $F(1, 57) = 2.99$, $p = .089$, $\eta^2 = 0.04$. It seems that the beneficial effect of retrieval practice in the story condition is less strong than in the word pairs condition.

The mixed design ANCOVA on the multiple choice test showed that the scores on the covariate, the vocabulary test, were significantly related to the scores on the multiple choice test, $F(1, 57) = 34.02$, $p < .001$, $\eta^2 = 0.31$. On the final multiple choice test, there was no significant main effect of learning condition after controlling for the effect of vocabulary size, $F < 1$. There was a significant main effect of context after controlling for the effect of vocabulary size, $F(1, 57) = 18.46$, $p < .001$, $\eta^2 = 0.17$, indicating that children in the word pairs condition recognized more synonyms than children in the story condition (word pairs: $M = 0.83$, $SD = 0.14$; versus story: $M = 0.72$, $SD = 0.16$). There was no interaction between learning condition and context, $F < 1$.

During scoring of the final cued recall test, we noticed that many children had produced synonyms that were incorrect but semantically similar to the synonyms presented in the experiment. Thus, we performed an additional analysis in which we applied a more liberal scoring method, counting semantically similar synonyms as correct. Using this more liberal scoring method, we obtained the same pattern of results as with the more strict scoring method.

Furthermore, we did an additional analysis on the final cued recall test in which we excluded the words that the children already knew at the pretest from the analysis. Again, we obtained the same pattern of results as in the scoring method in which also known words were included (Table 5).

Table 3. Proportion correct on the cued recall test in the word pairs condition and story condition for restudied and retrieved words (*SD* in parentheses)

	Condition	
Final test	Word Pairs ($n = 30$)	Story ($n = 30$)
Restudied words	0.41 (0.22)	0.36 (0.20)
Retrieval practice words	0.53 (0.23)	0.41 (0.20)

Table 4. Proportion correct on the multiple choice test in the word pairs condition and story condition for restudied and retrieved words (*SD* in parentheses)

	Condition	
Final test	Word pairs ($n = 30$)	Story ($n = 30$)
Restudied words	0.82 (0.15)	0.71 (0.22)
Retrieval practice words	0.83 (0.16)	0.73 (0.18)

Table 5. Proportion correct on the cued recall test in the word pairs condition and story condition for restudied and retrieved words (*SD* in parentheses) when words already known at the pretest were excluded from the analysis

	Condition	
Final test	Word pairs ($n = 30$)	Story ($n = 30$)
Restudied words	0.40 (0.22)	0.36 (0.20)
Retrieval practice words	0.53 (0.23)	0.40 (0.20)

DISCUSSION

Our main research question was whether retrieval practice benefits vocabulary learning in primary school children. The results of the final cued recall test showed that there

[1] The careful reader might note that the different values of η^2 sum up to more than 1. However, this occurs because effect sizes are calculated separately for the between-subjects variables and the within-subjects variables.

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was a benefit of repeated retrieval practice on the long-term when compared with repeated study. Children recalled more word synonyms that they had retrieved during learning than word synonyms that they had restudied. To our knowledge, this is the first study that showed a benefit of retrieval practice in primary school vocabulary learning. This study extends the earlier findings from adult vocabulary learning regarding the positive effect of retrieval practice (e.g. Carpenter et al., 2008; Karpicke & Smith, 2012) to primary school vocabulary learning. The positive effect of retrieval practice that we found in a classroom-based setting with current learning material is in line with recent classroom experiments on text learning (e.g. Butler & Roediger, 2007; McDaniel, Agarwal, Huelser, McDermott, & Roediger, 2011; McDaniel, Anderson, Derbish, & Morrisette, 2007; Roediger, Agarwal, McDaniel, & McDermott, 2011). In each of these studies, the to-be-learned materials were embedded in some kind of meaningful context. However, to the best of our knowledge, none of these studies aimed at investigating whether the retrieval practice effect varies with context.

The second question was whether providing a context would affect the benefits of retrieval practice. Our results suggest that it does as we found a marginally significant interaction between learning condition and context. This interaction effect showed that the positive effect of retrieval practice in the word pairs condition was somewhat larger than in the story condition. However, children in the word pairs condition retrieved more words than the children in the story condition during the first retrieval practice sessions. This in turn might be a plausible explanation for the difference in the retrieval practice effects between the word pairs condition and the context condition. All in all, because the interaction between learning condition and context was just marginally significant, we have to be careful with interpreting the data.

One remaining question is why we did not find any differences between restudying and retrieval practice on the multiple choice test. Although we did not expect these results, these results are in line with some other studies in which also no benefit of retrieval practice was found on recognition tasks (e.g. Hogan & Kintsch, 1971; Verkoeijen, Tabbers, & Verhage, 2011; Wenger, Thompson, & Bartling, 1980). However, our experiment was different from these studies in the sense that in our experiment, the recognition test was always preceded by the final cued recall test, which may have confounded the effect of learning condition on recognition performance. Therefore, we think it is better to base our claims about the beneficial effect of retrieval practice on the final cued recall test given before the multiple choice test.

Another remaining question is why providing a context of a story did not lead to a memory benefit compared with providing word pairs. Certainly, we did not expect a better performance on both the cued recall test and the multiple choice test for the word pairs condition than for the story condition. It should be noted, however, that the results are consistent with other studies in which no benefit of adding contextual information was found in vocabulary learning (e.g. Jones, Levin, Levin, & Beitzel, 2000; McDaniel & Pressley, 1984, 1989). For example, the results of the study of McDaniel and Pressley (1984) with graduate students showed that a context method led to worse recall of definitions than a keyword method and a control method in which a one-word to two-word definition was given. Furthermore, Jones et al. (2000) found a benefit in recall of definitions of a mnemonic keyword strategy compared with a context strategy in sixth grade children. In contrast to the aforementioned studies (Jones et al., 2000; McDaniel & Pressley, 1984, 1989), Rodríguez and Sadoski (2000) found a benefit of adding contextual cues to the keyword mnemonic in ninth grade children who had to learn Spanish-English word pairs. The combination method was superior to the keyword method, which contrasts with our results in which adding a context harmed recall of the words.

There are several possible explanations why we did not find a benefit of context in our study. One explanation is that the context was only provided in the first learning phase but not in the other learning phases and in the test session. Thus, a possible benefit of providing a context may have been diluted over time. However, in the retrieval practice trials that immediately followed the context presentation, words in the word pairs condition were already recalled better than words in the story condition. Thus, we do not think the dilution explanation is very plausible.

Another explanation may be that the contextual information diverted the children from the meaning of the words and disrupted the learning of the new word and its synonym. The explanation of the words throughout reading the story could have harmed the context benefit in the story condition. This fits well with the finding that words in the story condition were already remembered more poorly than in the word pairs condition in the first retrieval practice session. Also, the form of the cued recall test may have matched better with the word pairs condition than with the story condition. In the word pairs condition, the words were always shown without the story, which is identical to the presentation format in the final test. In contrast, in the story condition, the words were shown within the context of a story.

For future research, it would be interesting to address the benefit of retrieval practice in which context is also used in the other learning phases and in the final test phase, because the use of context is common in primary school vocabulary learning. In this way, we could better match the different learning phases and the final test phase with each other.

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REFERENCES

Anderson, R. C., & Freebody, P. (1981). Vocabulary knowledge. In J. T. Guthrie (Ed.), *Comprehension and teaching: Research reviews* (pp. 71–117). Newark, DE: International Reading Association.

Baker, S. K., Simmons, D. C., & Kame'enui, E. J. (1998). Vocabulary acquisition: Research bases. In D. C. Simmons, & E. J. Kame'enui

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(Eds.), *What reading research tells us about children with diverse learning needs* (pp. 183–218). Mahwah, NJ: Erlbaum.

Biemiller, A. (2005). Size and sequence in vocabulary development: Implications for choosing words for primary grade vocabulary instruction. In A. Hiebert, & M. Kamil (Eds.), *Teaching and learning vocabulary: Bringing research to practice* (pp. 223–242). Mahwah, NJ: Erlbaum.

Biemiller, A., & Boote, C. (2006). An effective method for building vocabulary in primary grades. *Journal of Educational Psychology*, *98*, 44–62. DOI: 10.1037/0022-0663.98.1.44

Blachowicz, C. L., Fisher, P. J., Ogle, D., & Watts-Taffe, S. (2006). Vocabulary: Questions from the classroom. *Reading Research Quarterly*, *41*, 524–530. DOI: 10.1598/RRQ.41.4.5

Bouwmeester, S., & Verkoeijen, P. P. J. L. (2011). Why do some children benefit more from testing than others? Gist trace processing to explain the testing effect. *Journal of Memory and Language*, *65*, 32–41. DOI: 10.1016/j.jml.2011.02.005

Butler, A. C., & Roediger, H. L. (2007). Testing improves long-term retention in a simulated classroom setting. *European Journal of Cognitive Psychology*, *19*, 514–527. DOI: 10.1080/09541440701326097

Carpenter, S. K., Pashler, H., & Vul, E. (2006). What types of learning are enhanced by a cued recall test? *Psychonomic Bulletin & Review*, *13*, 826–830. DOI: 10.3758/BF03194004

Carpenter, S. K., Pashler, H., Wixted, J. T., & Vul, E. (2008). The effects of tests on learning and forgetting. *Memory & Cognition*, *36*, 438–448. DOI: 10.3758/MC.36.2.438

Carrier, M., & Pashler, H. (1992). The influence of retrieval on retention. *Memory & Cognition*, *20*, 633–642. DOI: 10.3758/BF03202713

Cull, W. L. (2000). Untangling the benefits of multiple study opportunities and repeated testing for cued recall. *Applied Cognitive Psychology*, *14*, 215–235. DOI: 10.1002/(SICI)1099-0720(200005/06)14:3<215::AID-ACP640>3.0.CO;2–1

Deese, J. (1959). On the prediction of occurrence of particular verbal intrusions in immediate recall. *Journal of Experimental Psychology*, *58*, 17–22. DOI: 10.1037/h0046671

Delaney, P. F., Verkoeijen, P. P. J. L., & Spirgel, A. S. (2010). Spacing and testing effects: A deeply critical, lengthy, and at times discursive review of the literature. *Psychology of Learning and Motivation*, *53*, 63–147. DOI: 10.1016/S0079-7421(10)53003-2

Fritz, C. O., Morris, P. E., Nolan, D., & Singleton, J. (2007). Expanding retrieval practice: An effective aid to preschool children's learning. *The Quarterly Journal of Experimental Psychology*, *60*, 991–1004. DOI: 10.1080/17470210600823595

Gipe, J. P. (1979). Investigating techniques for teaching word meanings. *Reading Research Quarterly*, *14*, 624–644.

Hart, B., & Risley, T. (1995). *Meaningful differences in the everyday lives of young American children*. Baltimore: Paul H. Brookes.

Hogan, R. M., & Kintsch, W. (1971). Differential effects of study and test trials on long-term recognition and recall. *Journal of Verbal Learning and Verbal Behavior*, *10*, 562–567. DOI: 10.1016/S0022-5371(71)80029-4

Janssen, M., & Van Ooijen, M. (Eds.), (2012). *Taal Actief, 4, 5, 6, 7, 8*. Den Bosch: Malmberg.

Jones, M. S., Levin, M. E., Levin, J. R., & Beitzel, B. D. (2000). Can vocabulary-learning strategies and pair-learning formats be profitably combined? *Journal of Educational Psychology*, *92*, 256–262. DOI: 10.1037/0022-0663.92.2.256

Karpicke, J. D. (2009). Metacognitive control and strategy selection: Deciding to practice retrieval during learning. *Journal of Experimental Psychology: General*, *138*, 469–486. DOI: 10.1037/a0017341

Karpicke, J. D., & Roediger, H. L. (2008). The critical importance of retrieval for learning. *Science*, *319*, 966–968. DOI: 10.1126/science.1152408

Karpicke, J. D., & Smith, M. A. (2012). Separate mnemonic effects of retrieval practice and elaborative encoding. *Journal of Memory and Language*, *67*, 17–29. DOI: 10.1016/j.jml.2012.02.004

Marsh, E. J., Fazio, L. K., & Goswick, A. E. (2012). Memorial consequences of testing school-aged children. *Memory*, *20*, 899–906. DOI: 10.1080/09658211.2012.708757

McDaniel, M. A., Agarwal, P. K., Huelser, B. J., McDermott, K. B., & Roediger, H. L. (2011). Test-enhanced learning in a middle school science classroom: The effects of quiz frequency and placement. *Journal of Educational Psychology*, *103*, 399–414. DOI: 10.1037/a0021782

McDaniel, M. A., Anderson, J. L., Derbish, M. H., & Morrisette, N. (2007). Testing the testing effect in the classroom. *European Journal of Cognitive Psychology*, *19*, 494–513. DOI: 10.1080/09541440701326154

McDaniel, M. A., Howard, D. C., & Einstein, G. O. (2009). The read-recite-review study strategy: Effective and portable. *Psychological Science*, *20*, 516–522. DOI: 10.1111/j.1467-9280.2009.02325.x

McDaniel, M. A., & Pressley, M. (1984). Putting the keyword method in context. *Journal of Educational Psychology*, *76*, 598–609. DOI: 10.1037/0022-0663.76.4.598

McDaniel, M. A., & Pressley, M. (1989). Keyword and context instruction of new vocabulary meanings: Effects on text comprehension and memory. *Journal of Educational Psychology*, *81*, 204–213. DOI: 10.1037/0022-0663.81.2.204

Nation, I. S. P. (2001). *Learning vocabulary in another language*. Cambridge: Cambridge University Press.

Pashler, H., Cepeda, N. J., Wixted, J. T., & Rohrer, D. (2005). When does feedback facilitate learning of words? *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *31*, 3–8. DOI: 10.1037/0278-7393.31.1.3

Pyc, M. A., & Rawson, K. A. (2007). Examining the efficiency of schedules of distributed retrieval practice. *Memory & Cognition*, *35*, 1917–1927. DOI: 10.3758/BF03192925

Pyc, M. A., & Rawson, K. A. (2009). Testing the retrieval effort hypothesis: Does greater difficulty correctly recalling information lead to higher levels of memory? *Journal of Memory and Language*, *60*, 437–447. DOI: 10.1016/j.jml.2009.01.004

Pyc, M. A., & Rawson, K. A. (2011). Costs and benefits of dropout schedules of test-restudy practice: Implications for student learning. *Applied Cognitive Psychology*, *25*, 87–95. DOI: 10.1002/acp.1646

Rodríguez, M., & Sadoski, M. (2000). Effects of rote, context, keyword, and context/keyword methods on retention of vocabulary in EFL classrooms. *Language Learning*, *50*, 385 412. DOI: 10.1111/0023-8333.00121

Roediger, H. L., Agarwal, P. K., McDaniel, M. A., & McDermott, K. B. (2011). Test enhanced learning in the classroom: Long-term improvements from quizzing. *Journal of Experimental Psychology: Applied*, *17*, 382–395. DOI: 10.1037/a0026252

Roediger, H. L., & Karpicke, J. D. (2006). The power of testing memory: Basic research and implications for educational practice. *Perspectives on Psychological Science*, *1*, 181–210. DOI: 10.1111/j.1745-6916.2006.00012.x

Roediger, H. L., & McDermott, K. B. (1995). Creating false memories: Remembering words not presented on lists. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *21*, 803–814. DOI: 10.1037/0278-7393.21.4.803

Rohrer, D., Taylor, K., & Sholar, B. (2010). Tests enhance the transfer of learning. *Journal of Experimental Psychology: Learning, Memory & Cognition*, *36*, 233–239. DOI: 10.1037/a0017678

Schrooten, W., & Vermeer, A. (1994). *Woorden in het basisonderwijs. 15.000 woorden aangeboden aan leerlingen. [Words in primary education. 15,000 words offered to pupils]*. Tilburg: Tilburg University Press.

Stahl, S. A., & Fairbanks, M. M. (1986). The effects of vocabulary instruction: A model based meta-analysis. *Review of Educational Research*, *56*, 72–110. DOI: 10.3102/00346543056001072

Toppino, T. C., & Cohen, M. S. (2009). The testing effect and the retention interval: Questions and answers. *Experimental Psychology*, *56*, 252–257. DOI: 10.1027/1618-3169.56.4.252

Tulving, E. (1967). The effects of presentation and recall of material in free-recall learning. *Journal of Verbal Learning and Verbal Behavior*, *6*, 175–185. DOI: 10.1016/S0022-5371(67)80092-6

Van de Gein, J., Van de Guchte, C., & Kouwenberg, B. (2008). *Zin in Taal Nieuw A, B, C, D, E*. Tilburg: Zwijsen.

Verhoeven, L., & Vermeer, A. (1993). *Taaltoets allochtone kinderen bovenbouw. [Language proficiency test for ethnic minority children in grades 3 to 6]*. Tilburg: Zwijsen.

Verkoeijen, P. P. J. L., Tabbers, H. K., & Verhage, M. L. (2011). Comparing the effects of testing and restudying on recollection in recognition memory. *Experimental Psychology*, *58*, 490–498. DOI: 10.1027/1618-3169/a000117

Vermeer, A. (2000). Lexicale rijkdom, tekstmoeilijkheid en woordenschatgrootte. Beschrijving van de MLR, een woordenschat-analyseprogramma. *Toegepaste Taalwetenschap in Artikelen*, *64*, 95–105.

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Vermeer, A. (2001). Breadth and depth of vocabulary in relation to acquisition and frequency of input. *Applied Psycholinguistics*, *22*, 217–234. DOI: 10.1017/S0142716401002041

Webb, L. W. (1921). A comparison of two methods of studying with application to foreign language. *The School Review: A Journal of Secondary Education*, *29*, 58–67.

Wenger, S. K., Thompson, C. P., & Bartling, C. A. (1980). Recall facilitates subsequent recognition. *Journal of Experimental Psychology: Human Learning and Memory*, *6*, 135–144. DOI: 10.1037/0278-7393.6.2.135

Wheeler, M. A., Ewers, M., & Buonanno, J. F. (2003). Different rates of forgetting following study versus test trials. *Memory*, *11*, 571–580. DOI: 10.1080/09658210244000414

APPENDIX A

Story in Dutch with 20 difficult words for the story condition

Ayoena, een meisje van 10 jaar oud, gaat op bezoek bij haar grootmoeder. Wanneer ze haar huis inloopt, roept ze: 'Het is een **chaos** hier!'. 'Wat is dit?', roept ze dan **kwiek**. Ze komt terug met een **vaal** tasje in haar hand. Het is helemaal versleten en **weerzinwekkend**. Een brede klep met riempjes sluit het tasje af. 'Oma, wat is dit? Het lag bij opa's oude **baret**.' 'Vast niets', zegt oma. Misschien moeten we dat oude tasje maar in de vuilniszak deponeren,' zegt oma.

Ayoena wil niet dat het tasje in de vuilniszak gaat, want het is een heel **apart** tasje. Ayoena peutert **heimelijk** de **stugge** riempjes van het tasje los en doet de klep open. 'Er zit iets in!' Oma kijkt **perplex** als ze een **kris** tevoorschijn haalt.

'De kris van opa,' fluistert oma Ietje. Ik dacht dat hij hem al lang weggegeven had. Dit was echt opa's **pronkstuk**'. Ze denkt terug aan opa die **heengegaan** is en begint te wenen. Ayoena denkt dat oma verdriet heeft om haar. Ayoena denkt dat ze het **verprutst** heeft, maar dat is niet zo. Oma is eigenlijk ook wel blij met het **signaal** van opa.

Beduusd roept Ayoena: 'Er zit een briefje bij de kris!' Ayoena vouwt het briefje open en leest hardop: 'Ik heb het **vermoeden** dat mijn nieuwsgierige kleindochter nu mijn kris gevonden heeft. Als dat het geval is, is hij voor haar bestemd. Ik hoop dat je deze laatste **gift** aanneemt, Ayoena. Oma zal je **meedelen** hoe je met een kris moet omgaan. Dag. Opa.'

APPENDIX B

English translation of story in Dutch with 20 difficult words for the story condition

Ayoena, a girl of 10 years old is visiting her grandmother. When she walks into the house, she shouts 'It is total **chaos** here!' Then she shouts **briskly**, 'What is this?' She has a **colourless** bag in her hand. It is totally worn out and **repulsive**. You can close the bag with a broad flap with belts. 'Grandmother, what is this? It was lying next to the old **beret** of grandfather.' 'Surely nothing', grandmother says. 'Maybe we have to **dispose** the old bag in the trashcan'.

Ayoena does not want to put the bag in the trashcan, because it is a very **unique** bag. Ayoena tampers **secretly** with the **rigid** belts and opens the flap. 'There is something in it!' Grandmother looks **speechless** when she shows **a dagger**.

Grandmother whispers, 'Grandfather's dagger.' I thought he had already given it away. This really was grandfather's **showpiece**'. She thinks back to grandfather who **passed away** and starts to **weep**. Ayoena thinks grandmother is crying because of her. Ayoena thinks she has **failed**, but that is not true. Actually, grandmother is very happy with **the sign** of grandfather.

Ayoena shouts **perplexed**: 'There is a note in the bag with the dagger!' Ayoena opens the note and reads aloud: 'I have the **assumption** that my curious granddaughter has found my dagger. If this is the case, it is for her. I hope you will accept this last **offering**, Ayoena. Grandmother will **describe** to you how to use the dagger. Goodbye. Grandfather.'

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